Course Outline for CSE-347

Part A

1. Course Code: CSE-347

2. Course Title: Introduction to Mathematical Programming

3. Course Type: Core Course

4. Level/ Term: Level: 3 Term: II

5. Academic Session: 2020-21

6. Course Teacher: Ashim Dey, Assistant Professor, CSE, CUET
and Saadman Sakib, CSE, CUET

7. Prerequisite(s): None

8. Credits: 3

9. Contact Hours: 3 lectures of 50 minutes per week

10. Total Marks: 300

11. Rationale of the Course:

This course will be an introduction to mathematical programming, with an emphasis on techniques for the solution and analysis of deterministic linear models. The primary types of models to be addressed will be linear programming, network flow, and integer linear programming. However, the course will touch on more complex models, such as those incorporating nonlinear constraints or uncertainty. The main emphasis will be on solution techniques and on analysis of the underlying mathematical structure of these models. In addition, this course will look at applications of optimization techniques in areas such as scheduling, routing, budgeting and resource allocation. This is a required course for all the students enrolling B. Sc. Engg. in CSE program. The catalogue description of the course is

Course Content:

Concepts of linear programming, preliminary theory and geometry of linear programming, basic feasible solution, complexity of linear programming; Simplex method, variants of simplex method, and revised simplex method, complexity of simplex method; Duality and its principles, interpretation of dual variables, dual simplex method, primal-dual method; Linear integer programs and their applications in real decision making problems; Sensitivity analysis; Large scale optimization; Transportation problems, Assignment problems, Network maximum flow problems; The ellipsoid method; Interior point methods; Integer programming methods; Nonlinear programming; Lagrange multipliers; Linear optimization; Discrete optimization; Dynamic programming; Engineering applications of optimization; Queuing models: M/M/1, M/M/C, M/G/1, M/D/1, G/M/1 solution of network of queue-closed queuing models, approximate solution methods, Application of queuing models in Computer Science.

12. Course Objectives:

a) To improve the ability to rigorously prove mathematical statements.
b) To develop knowledge of the mathematical structure of the most commonly used deterministic linear optimization models.
c) To analyze the structures and mathematical models of various complex system occurring in real world applications.
d) To be able to apply various algorithms to solve integer programming problems.

13. Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs)

a) CLOs

	Course Learning Outcomes (CLOs)	**Blooms Level (Optional)**
CLO1	Apply the knowledge of linear and non-linear programming to solve complex optimization problems.	C3
CLO2	Assess when optimization models might be solved by exact methods and when heuristics are needed.	C3, C6
CLO3	Design solutions to real world problems using the concepts of various queuing models.	C5, P7

b) Mapping of CLO with PLO

No.	(CLOs)	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
1	CLO1			X									
2	CLO2		X										
3	CLO3			X									

Part B

14. Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs

	Topic	Teaching-Learning Methodology	Assessment Method	Corresponding CLOs
Week-01	Introduction to linear programming	• Lecture • Presentation slide	Class Test Final Exam	CLO-1
Week -02	Simplex methods and its variants	• Lecture • Presentation slide	Class Test Final Exam	CLO-1
Week -03	Duality principles	• Lecture • Presentation slide	Class Test Final Exam Assignments	CLO-1
Week -04	Dual simplex method and primal dual methods	• Lecture • Presentation slide	Class Test Final Exam	CLO-1
Week -05	Linear integer programs and their applications	• Lecture	Class Test Final Exam Assignments	CLO-1

		• Presentation slide		
Week -06	Sensitivity analysis, large scale optimization	• Lecture • Presentation slide	Class Test Final Exam Assignments	CLO-2
Week -07	Transportation problem and assignment problem	• Lecture • Presentation slide	Class Test Final Exam Assignments	CLO-2
Week -08	Network maximum flow	• Lecture • Presentation slide	Class Test Final Exam	CLO-3
Week -09	Ellipsoid method and interior point method	• Lecture • Presentation slide	Class Test Final Exam Assignments	CLO-3
Week -10	Non-linear programming	• Lecture • Presentation slide	Class Test Final Exam Assignments	CLO-3
Week -11	Discrete optimization	• Lecture • Presentation slide	Class Test Final Exam Assignments	CLO-3
Week -12	Queuing models and its applications	• Lecture • Presentation slide	Final Exam	CLO-2
Week -13	Review and makeup classes	• Lecture	Final Exam	CLO-2

Part C

15. Assessment and Evaluation

l) Assessment Strategy

Class participation and attendance	10%
Class tests / Class assessment	20%
Term Final Examination (3 hours duration)	70%
Total	100%

2) Marks distribution:

a) Continuous Assessment: 30%

b) Summative: 70%

c) Make-up Procedures:

- Feedback on continuous assessment is given to the students immediately after the test.
- The minimum number of class-test/assignments are (n+1) with best n will be counted (here, n is number of credit). Based on the students' feedback additional class-test/assignment may be taken by the course teacher

Part D

16. Learning Materials

1) Required Textbook:

- Book: ***Introduction to Mathematical Programming: Operations Research***, 4th edition by Wayne L. Winston, Munirpallam Venkataramanan, ISBN-13: 978-0534359645

2) Reference Book:

- Book: ***Introduction to Linear Optimization*** by Dimitris Bertsimas and John N. Tsitsiklis, Athena Scientific, ISBN-13: 978-1886529199